# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The magnetic field lines outside a bar magnet are directed 1

**(a)** from the north pole to the south pole **(b)** in no definite direction

**(c)** along the magnet only **(d)** from the south pole to the north pole

**2.** The field pattern of a solenoid is similar to that of 1

**(a)** a straight wire **(b)** a horseshoe magnet only

**(c)** a plane mirror **(d)** a bar magnet

**3.** Electromagnetic induction is the production of 1

**(a)** induced current in a coil due to a changing magnetic field

**(b)** a magnetic field by a steady current

**(c)** force on a coil

**(d)** heat by current

**4.** Magnetic field lines inside a bar magnet go from 1

**(a)** the north pole to the south pole **(b)** the south pole to the north pole

**(c)** do not exist **(d)** one side to the other randomly

**5.** A soft iron core inside a solenoid 1

**(a)** reduces the field **(b)** makes an electromagnet

**(c)** stops the current **(d)** has no effect

**6.** The phenomenon of electromagnetic induction was discovered by 1

**(a)** Ampere **(b)** Michael Faraday **(c)** Newton **(d)** Oersted

**7.** Two magnetic field lines 1

**(a)** intersect at the poles **(b)** intersect often

**(c)** never intersect each other **(d)** intersect at the centre

**8.** An electromagnet is made of 1

**(a)** plastic

**(b)** copper only

**(c)** soft iron wrapped with a coil carrying current

**(d)** steel only

**9.** An induced current is produced in a coil when 1

**(a)** a magnet is held still near it
**(b)** the coil is kept in a steady field at rest
**(c)** a magnet is moved towards or away from it
**(d)** a battery is disconnected

**10.** The strength of a magnetic field is indicated by 1

**(a)** the colour of the lines **(b)** the number of poles
**(c)** the closeness of the field lines **(d)** the length of the lines

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** State the rule to find the direction of the magnetic field around a straight current-carrying wire. 2

**12.** Name the rule to find the direction of the force on a current-carrying conductor and state it. 2

**13.** State two advantages of AC over DC. 2

**14.** List any two properties of magnetic field lines. 2

## SECTION — C (Short Answer Type) 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Draw (describe) the magnetic field lines around a bar magnet and state three properties. 3

**16.** A proton enters a magnetic field perpendicular to it. State what happens and the direction of force. 3

**17.** What are the colour codes of the live, neutral and earth wires? 3

OR

Describe Oersted's experiment and state the conclusion. 3

**18.** State Fleming's left-hand rule with the help of the three fingers. 3

OR

How does a fuse protect the circuit from overloading? 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Describe an activity to show that a current-carrying wire produces a magnetic field around it. State the right-hand thumb rule. 5

OR

Draw (describe) a domestic electric circuit and explain the functions of the live wire, neutral wire, earth wire and fuse. 5

**20.** Describe the magnetic field due to a current-carrying solenoid with the field pattern. Explain how it is used to make an electromagnet. 5

OR

What are short circuiting and overloading? State their causes and the safety devices used. 5

## END OF QUESTION PAPER Total: 40 Marks